Code: 20A02601T

R20

B.Tech III Year II Semester (R20) Regular Examinations August 2023

POWER SYSTEM ANALYSIS

(Electrical & Electronics Engineering)

Time: 3 hours | Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

	(a)	What is the advantage of per unit method over percent method?	2M
	(b)	Define base impedance and base kilovolt amperes.	2M
	(c)	What is meant by partial network?	2M
	(d)	What are the methods available for forming bus impedances matrix?	2M
	(e)	What is the need for load flow study?	2M
	(f)	Define voltage controlled bus.	2M
	(g)	What is the significance of operator ‘a’?	2M
	(h)	Write the symmetrical components of three phase system.	2M
	(i)	What are the factors that affect the transient stability?	2M
	(j)	Define critical clearing angle.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2	(a)	Why do you use a single line diagram for power system representation discuss with example?	5M
	(b)	Explain the relationship between: (i) The basic loops and links. (ii) Basic cut-sets and the number of branches.	5M
		OR	
3		Derive static load flow equations of a power system and mention the assumptions made.	10M
4		Explain the algorithm for the addition and removal of lines in power system.	10M
		OR	
5		Describe the construction of Bus impedance matrix using building algorithm for line without mutual coupling.	10M
6	(a)	What is acceleration factor? What is its role in GS-method for power flow studies?	5M
	(b)	Differentiate between G-S and N-R Power flow methods.	5M
		OR	
7		Develop an algorithm for N-R method in polar coordinate method.	10M
8	(a)	Derive the expression for 3-Phase power in terms of symmetrical components.	5M
	(b)	Explain the importance of sequence impedances of an unloaded synchronous generator.	5M
		OR	
9		Derive the necessary equation to determine the fault current for Single Line Ground fault and draw the connection diagram off its sequence networks.	10M
10		Discuss the any two methods to improve steady state stability.	10M
		OR	
11		A generator having H = 16.0 MJ / MVA is delivering 1.0 pu to an infinite bus via a purely reactive network. When occurrence of a fault reduces the generator output power to zero? The maximum power that can be delivered is 6.5 pu. When the fault is cleared the original network conditions again exists? Determine critical clearing angle and critical clearing time.	10M

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